

Chapter 5: Far and Near

Worksheet 5: Far and Near

Learning Objective: Measure and estimate length, convert between mm, cm, m and km, and compare real-life distances.

Worked Example:

A pencil is 14 cm long and a pen is 126 mm long. Which is longer, and by how much (in cm)?

Solution: $126 \text{ mm} = 12.6 \text{ cm}$. The pencil (14 cm) is longer by $14 - 12.6 = 1.4 \text{ cm}$.

Questions (1–20)

1. How many millimetres make 1 centimetre?
2. How many metres make 1 kilometre?
3. Convert 250 cm into metres.
4. Convert 3 km into metres.
5. Which unit would you use to measure the length of a football field: cm, m or km?
6. A rope is 4 m 25 cm long. Express this length in centimetres only.

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7. The distance between two villages is 3,500 m. Express this in km.

 8. Ramesh's height is 142 cm. Express his height in metres and centimetres.

 9. A table is 1.2 m long. Express its length in cm.

 10. Order these lengths from shortest to longest: 350 cm, 3 m, 3,500 mm, 0.3 km.

 11. Asha measures the classroom length as 8 m 40 cm; Vinod measures it as 835 cm. Whose measurement is longer, and by how much?

 12. A road is built in three sections: 1.2 km, 850 m and 2.05 km. Find the total length of the road in metres.

 13. A wire of length 5 m is cut into 4 equal pieces. Find the length of each piece in cm.

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14. The distance from home to school is 1 km 250 m. A student walks this distance twice a day (to and from school). Find the total distance walked in a day, in metres.
15. A ladder is 350 cm long. Will it fit through a doorway that is 3.2 m tall? Explain.
16. A cyclist travels 2.5 km to the market, then 1,750 m to a friend's house, and finally 800 m back home. Find the total distance travelled, in km.
17. A tailor has a 10 m cloth roll and cuts pieces of 1 m 25 cm each. How many full pieces can he cut, and how much cloth is left over?
18. A school ground is 120 m long and 80 m wide. If a student jogs around the boundary (perimeter) twice, how much distance is covered?
19. Measure the length of your study table or bed using a tape (or estimate using steps or handspans) and write the length in both cm and m.

20. Estimate the distance from your home to the nearest market or place of worship. Write your estimate in km or m, and explain how you estimated it.

Reflection Question: Which is harder for you to imagine – a length in millimetres or a length in kilometres? Why do you think different units are used for different things?

Real-Life Application Question: Look at a road sign near your home that shows the distance to another town or city. Write down the distance and the unit used.